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March 14, 2025

Faisal D'Souza
NCO/NITRD
2415 Eisenhower Avenue
Alexandria, VA 22314

Dear Mr. D'Souza,

Thank you for the opportunity to provide comments on the development of a national Artificial Intelligence (AI) Action Plan. As one of the leading healthcare organizations in the world, we appreciate the opportunity to respond to this RFI.

Duke Health integrates the Duke University Health System (all of our hospitals, clinics, and the Duke Health Integrated Practice), Duke University School of Medicine, Duke-NUS Medical School, Duke University School of Nursing, and incorporates the health and health research programs within the Duke Global Health Institute as well as those in schools and centers across Duke University, including the Duke Robert J. Margolis Center for Health Policy.

Our organization is committed to excellence through the adoption of AI innovations to improve patient outcomes, reduce administrative burden, and streamline care delivery. To achieve this, we established the Algorithm-Based Clinical Decision Support (ABCDS) Oversight Committee in 2020. This committee governs and reviews AI solutions for patient care. At that time, AI quality and ethical considerations were not widely understood or adopted by stakeholders, and FDA guidance was still under development. Over the past four years, we have worked to balance innovation with high-quality, ethical care, ensuring responsible deployment of AI solutions.

Based on our experience over the past four years, Duke Health offers the following considerations for the AI Action Plan directed by the Presidential Executive Order of January 23, 2025:

Risk-Based Approach to AI Assurance and Review
Duke Health employs a risk-based approach to evaluate AI tools, tailoring the required evidence to the potential benefits and risks within the context of use. High-risk AI solutions, such as those posing critical safety risks (e.g., death), require extensive evidence demonstrating adherence to ethical and quality principles, along with detailed monitoring plans. Conversely, low-risk solutions, like those reducing administrative burden (e.g. back office functions) with de-minimis safety implications, require less rigorous evidence. Core quality and ethical principles must be met for all AI tools, but a risk-based evaluation permits streamlined testing (local validation) with documented methods, results, and conclusions often leading to an expedited deployment, provided there's ongoing safety surveillance. This approach also ensures appropriate and efficient

allocation of resources throughout the evaluation process and the ongoing quality assurance efforts.

Transparency

AI solutions should be transparent. Developers should utilize methods like nutrition labels, model cards, and comprehensive documentation to ensure basic transparency about the solution itself, but also maintain intellectual property (IP) protections. This transparency includes detailing intended use, patient populations used for testing and training, model performance and outcomes, safety events, and other relevant issues, with contextual information about testing in specific settings and routine updates when changes occur. This transparency builds consumer confidence, and facilitates faster testing and adoption.

A contractual framework can streamline procurement and enhance accountability across all parties involved in the AI development and deployment process. Within the terms of the contracts, vendors should inform institutional customers if AI is being added or modified, and to share with customers how the new/modified AI has been tested and/or validated. Clear definition of roles and responsibilities in change management plans between developers and implementers, coupled with open communication about safety risks, performance, data use, and outcomes between implementers and developers, fosters continuous improvement and maximizes the potential of AI innovations.

Post-market surveillance and health AI outcomes registries

National health AI registries can empower AI solution developers to showcase the impact of their products on improving health outcomes. These registries would provide consumers and implementers with quick and easy access to information on tested and utilized AI solutions, including their benefits. This transparency fosters a competitive market focused on developing beneficial AI solutions. Furthermore, national registries can be seamlessly integrated with individual organizations' registries for efficient AI solution inventory management.

Voluntary use-case/tool registries can also provide a pathway to achieve local (user) accountability for AI safety and efficacy. We envision the development of public-private partnerships to share testing, implementation, evaluation best practices, impact/ROI results, and adverse events to further inform future development efforts.

Below are some editorials and articles that experts at Duke Health amongst others have published related to these topics:

- National registries: https://pubmed.ncbi.nlm.nih.gov/39133500/
- Registration information: https://academic.oup.com/jamia/article-abstract/31/3/705/7455696
- Model cards: https://chai.org/draft-chai-applied-model-card/ and https://www.nature.com/articles/s41746-020-0253-3
- The Regulation of Uncertainty: https://paragoninstitute.org/private-health/the-regulation-of-uncertainty/

Closing
Our phenomenal faculty, researchers, and physician scientists are some of the best in the world, and have served the Executive Branch and Congress as experts on a wide-range of topics. We hope you will utilize Duke Health as you think about the future of AI in healthcare.

Sincerely.

Jeffrey Ferranti, MD, MS Senior Vice President and Chief Digital Officer Duke Health	Michael J. Pencina, PhD Chief Data Scientist Duke Health